

Exposé XIII. Finiteness theorems for the Picard functor

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1. Bounded sheaves

Let k be an algebraically closed field, and let X be a projective k -scheme endowed with an ample invertible sheaf $\mathcal{O}_X(1)$.

Definition 1.1 (Mumford). Let $m \in \mathbb{Z}$. A coherent sheaf F on X is called *m -regular* if $\mathcal{O}_X(1)$ is generated by $H^0(\mathcal{O}_X(1))$ at every point of $\text{Supp}(F)$, and if, for every $q > 0$,

$$H^q(F(m - q)) = 0.$$

Lemma 1.2. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be an exact sequence of coherent sheaves on X , where the first arrow is multiplication by a section of $\mathcal{O}_X(1)$. If F is m -regular, then G is m -regular.

Indeed, the cohomology sequence contains

$$H^q(F(m - q)) \longrightarrow H^q(G(m - q)) \longrightarrow H^{q+1}(F(m - q - 1)),$$

and the two outside terms vanish by m -regularity.

Proposition 1.3. If F is m -regular, then, for every $n \geq m$,

- i) F is n -regular;
- ii) the multiplication map

$$H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) \longrightarrow H^0(F(n + 1))$$

is surjective;

- iii) $F(n)$ is generated by its global sections.

The proof is by induction on $s = \dim \text{Supp}(F)$. The case $s = -1$ is empty. Choose a section $\sigma \in H^0(\mathcal{O}_X(1))$ which vanishes at no point of the finite set $\text{Ass}(F)$. It gives an exact sequence

$$0 \longrightarrow F(-1) \xrightarrow{\sigma} F \longrightarrow G \longrightarrow 0,$$

and Lemma 1.2 shows that G is m -regular, with support of dimension $s - 1$. From

$$H^q(F(n - q - 1)) \longrightarrow H^q(F(n - q)) \longrightarrow H^q(G(n - q))$$

the induction hypothesis gives $H^q(G(n - q)) = 0$ for $n \geq m$; hence the vanishing for $F(n - q - 1)$ implies the vanishing for $F(n - q)$. Starting with $n = m$, this proves (i).

For (ii) one uses the multiplication diagram, whose rows come from the exact sequences above:

$$\begin{array}{ccccc} H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) & \longrightarrow & H^0(F(n+1)) & \longrightarrow & H^0(G(n+1)) \\ \sigma \otimes 1 \downarrow & & \downarrow \sigma & & \downarrow \sigma \\ H^0(F(n+1)) \otimes H^0(\mathcal{O}_X(1)) & \longrightarrow & H^0(F(n+2)) & \longrightarrow & H^0(G(n+2)). \end{array}$$

The first row is exact by (i), and the surjectivity needed for G is known by induction; this gives the surjectivity for F . For (iii), writing $M_X = M \otimes_k \mathcal{O}_X$, the maps

$$H^0(F(n))_X \longrightarrow F(n)$$

are compatible with multiplication by $H^0(\mathcal{O}_X(1))$. Once such a map is surjective for one n , (ii) gives it for all larger n ; Serre's theorem gives the assertion for $n \gg 0$, hence for every $n \geq m$.

Proposition 1.4. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be exact on X . If G is m -regular, then:

- i) $H^q(F(n)) = 0$ for $q \geq 2$ and $n \geq m - (q - 1)$;
- ii) the map $H^1(F(n-1)) \rightarrow H^1(F(n))$ is an isomorphism for

$$n \geq (m-1) + h^1(F(m-1));$$
- iii) $H^1(F(n)) = 0$ for the same range of n .

In particular F is $[(m-1) + h^1(F(m-1))]$ -regular.

Indeed, from

$$H^{q-1}(G(n)) \longrightarrow H^q(F(n-1)) \longrightarrow H^q(F(n)) \longrightarrow H^q(G(n))$$

the outside terms vanish for $q \geq 2$ and $n \geq m - (q - 1)$ by Proposition 1.3. Thus

$$H^q(F(m-q)) \simeq H^q(F(m-q+1)) \simeq H^q(F(m-q+2)) \simeq \cdots,$$

and the common value is zero by Serre's theorem. For $n \geq m - 1$ one has the exact row

$$0 \rightarrow H^0(F(n-1)) \rightarrow H^0(F(n)) \rightarrow H^0(G(n)) \rightarrow H^1(F(n-1)) \rightarrow H^1(F(n)) \rightarrow 0.$$

If $H^0(F(n)) \rightarrow H^0(G(n))$ is surjective, then the multiplication diagram of Proposition 1.3 gives the same surjectivity in degree $n+1$. If surjectivity has not yet occurred, $h^1(F(n-1))$ strictly decreases when passing to $h^1(F(n))$. There can be at most $h^1(F(m-1))$ such failures; this proves (ii) and (iii).

Definition 1.5. Let F be a coherent sheaf on X , let $r \geq \dim \text{Supp}(F)$, and let $(b) = (b_0, \dots, b_r)$ be a sequence of $r+1$ integers. We say that F is a (b) -sheaf if $\mathcal{O}_X(1)$ is generated by global sections on $\text{Supp}(F)$, if

$$h^0(F(-1)) \leq b_0,$$

and, for $r \geq 1$, if there exists a section $\sigma \in H^0(\mathcal{O}_X(1))$ giving an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

with G a $(b_1, \dots, b_r)$ -sheaf. Equivalently, there is an F -regular sequence of r sections $\sigma_1, \dots, \sigma_r$ of $H^0(\mathcal{O}_X(1))$ such that, if F_{σ_i} denotes the restriction of F to the common zero scheme of $\sigma_1, \dots, \sigma_i$, then

$$h^0(F_{\sigma_i}(-1)) \leq b_i \quad (0 \leq i \leq r).$$

Proposition 1.6. If F is a (b) -sheaf, then:

- i) for every sufficiently general sequence $\sigma_1, \dots, \sigma_r$ of sections of $H^0(\mathcal{O}_X(1))$, the restrictions F_{σ_i} satisfy $h^0(F_{\sigma_i}(-1)) \leq b_i$ for $0 \leq i \leq r$;
- ii) every subsheaf $G \subset F$ is again a (b) -sheaf.

The proof is the standard one: on the open set of regular sequences, the sheaves $F_{\sigma_i}(-1)$ form flat families; upper semicontinuity gives (i), and a sufficiently general sequence is also regular for any subsheaf G .

Lemma 1.7. If

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$$

is exact, and if

$$\chi(F(n)) = \sum_{i=0}^r a_i \binom{n+i}{i},$$

then

$$\chi(G(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

Indeed $\chi(G(n)) = \chi(F(n)) - \chi(F(n-1))$ and

$$\binom{n+i}{i} - \binom{n-1+i}{i} = \binom{n+i-1}{i-1}.$$

Proposition 1.8. Let F be a (b) -sheaf, put $s = \dim \text{Supp}(F)$, and write

$$\chi(F(n)) = \sum_{i=0}^s a_i \binom{n+i}{i}.$$

Then:

- i) for $n \geq -1$,

$$h^0(F(n)) \leq \sum_{i=0}^s b_i \binom{n+i}{i};$$

- ii) $a_i \leq b_i$ for every i , and F is also a $(b_0, \dots, b_{s-1}, a_s)$ -sheaf.

The proof is by induction on s . For $s = 0$ one has $a_0 = h^0(F) = h^0(F(-1)) \leq b_0$. For $s \geq 1$, choose an exact sequence $0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$ with G a $(b_1, \dots, b_s)$ -sheaf; then

$$h^0(F(n)) - h^0(F(n-1)) \leq h^0(G(n)),$$

and the assertion follows from the induction hypothesis and Lemma 1.7.

Definition 1.9. The (b) -polynomials are universal polynomials with non-negative rational coefficients

$$P_r(b_0, \dots, b_{r-1}; x),$$

defined recursively from the preceding regularity estimates: $P_0 = 0$, and P_r is obtained by applying Proposition 1.4 to a hyperplane section controlled by P_{r-1} . They are increasing in each argument.

Remark 1.10. Immediately by induction on $r \geq 1$,

$$P_r(b_0, \dots, b_{r-1}; P_{r-1}(b_1, \dots, b_{r-1}; 0)) = P_r(b_0, \dots, b_{r-1}; 0).$$

Theorem 1.11. Let F be a (b) -sheaf and write

$$\chi(F(n)) = \sum_{i=0}^s a_i \binom{n+i}{i}.$$

Let $(c) = (c_1, \dots, c_s)$ be a sequence with $(c) \geq (b) - (a)$, and put $m = P_s(c)$. Then $m \geq 0$ and F is m -regular. In particular, if $s = \dim \text{Supp}(F)$, then F is

$$P_s(a_0, \dots, a_{s-1}; a_s)$$

regular.

The proof is by induction on s . The case $s = 0$ is clear. For $s \geq 1$, choose $0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$ with G a $(b_1, \dots, b_s)$ -sheaf. By Lemma 1.7 and the induction hypothesis, G is regular with a bound controlled by P_{s-1} ; Proposition 1.4 then gives the asserted bound for F .

Definition 1.12. Let $X \rightarrow S$ be of finite type. A family $\mathcal{F}$ of classes of coherent sheaves on geometric fibres of X/S is a set of classes stable under extension of the ground field. It is *bounded* by a coherent sheaf F on $X_T = X \times_S T$, with T of finite type over S , if $\mathcal{F}$ is contained in the family of classes of the fibres $F_{k(t)}$. If X/S is projective with a relatively ample sheaf $\mathcal{O}_X(1)$, the family is a (b) -family if every class is represented, after extension to an algebraic closure, by a (b) -sheaf.

Theorem 1.13. Let S be noetherian and let X be projective over S , endowed with an ample invertible sheaf $\mathcal{O}_X(1)$ whose restrictions to the fibres are generated by global sections. For a family $\mathcal{F}$ of classes of coherent sheaves on the fibres of X/S , the following conditions are equivalent:

- i) $\mathcal{F}$ is bounded. Moreover, if all sheaves are locally free of rank p , the bounding sheaf may be chosen locally free of rank p .
- ii) The set of Hilbert polynomials $\chi(F_s(n))$ is finite, and $\mathcal{F}$ is a (b) -family for some (b) .
- iii) The set of Hilbert polynomials is finite, and there is an integer m such that all sheaves in $\mathcal{F}$ are m -regular.
- iv) The set of Hilbert polynomials is finite, and $\mathcal{F}$ is contained in the family of quotient classes of a coherent sheaf E_T on some X_T ; in fact one may take $T = S$ and $E = \mathcal{O}_X(-m)^{\oplus M}$.
- v) $\mathcal{F}$ is contained in the family of cokernel classes of homomorphisms $E'_T \rightarrow E_T$, where E' and E are coherent on some X_T ; in fact one may take

$$E' = \mathcal{O}_X(-m')^{\oplus M'}, \quad E = \mathcal{O}_X(-m)^{\oplus M}.$$

If $\mathcal{F}$ is bounded by F on X_T , stratifying T and using generic flatness makes F flat over T . Then only finitely many Hilbert polynomials occur, and the locally free rank is locally constant. By semicontinuity and the hypotheses on $\mathcal{O}_X(1)$, after a finite stratification there are regular sequences of global sections controlling the numbers $h^0(F_{\sigma_i, t}(-1))$; hence (i) implies (ii). The implication (ii) $\Rightarrow$ (iii) is Theorem 1.11. If (iii) holds, Proposition 1.3 gives surjections

$$\mathcal{O}_X(-m)^{\oplus M} \twoheadrightarrow F$$

with M the maximum of the finitely many values $\chi(F(m))$, hence (iv). From a quotient sequence $0 \rightarrow G_T \rightarrow E_T \rightarrow F_T \rightarrow 0$, boundedness for E_T and finiteness of Hilbert polynomials for F_T imply the same finiteness for G_T ; Proposition 1.6 controls the kernels, giving (v). Finally, morphisms $E'_T \rightarrow E_T$ can, after stratification and fixed presentations by sums of twists, be represented by the affine space

$$V((f_T)_* \mathcal{H}om(E', E)^*),$$

which carries a universal homomorphism and a universal cokernel. This proves (v) $\Rightarrow$ (i).

2. Several technical lemmas

2.1. Let k be a field and let X be proper over k . If $L_1, \dots, L_s$ are invertible sheaves on X , and if Y is a closed subscheme of dimension at most s , the intersection number

$$\langle c_1(L_1) \cdots c_1(L_s) \cdot Y \rangle$$

is the coefficient of $n_1 \cdots n_s$ in Snapper's polynomial

$$\chi(L_1^{\otimes n_1} \otimes \cdots \otimes L_s^{\otimes n_s} \otimes \mathcal{O}_Y).$$

2.2. From now on in this section, k is algebraically closed and X is integral, projective, of dimension r , and endowed with a very ample invertible sheaf $H = \mathcal{O}_X(1)$. For an invertible sheaf L on X , the integer

$$d(L) = \langle c_1(L) c_1(H)^{r-1} \rangle$$

is called the degree of L , relative to H . Write

$$\chi(L(n)) = \sum_{i=0}^r a_i \binom{n+i}{i}, \quad \chi(\mathcal{O}_X(n)) = \sum_{i=0}^r b_i \binom{n+i}{i}.$$

Lemma 2.3. One has $a_r = b_r = d(H)$. If $r \geq 1$, the next coefficient is read from the corresponding intersection product involving $c_1(L)$ and $c_1(H)^{r-2}$, equivalently by comparing the Hilbert polynomials of L and $L \otimes H^{\otimes m}$ for $m = 0, 1$.

Lemma 2.4. If $d(L) \leq 0$ and $H^0(L) \neq 0$, then $L \simeq \mathcal{O}_X$; hence $d(L) = 0$ and $h^0(L) = 1$. A non-zero section identifies L with $\mathcal{O}_X(Y)$ for an effective divisor Y ; if $Y > 0$, then $\langle c_1(Y) c_1(H)^{r-1} \rangle > 0$.

Lemma 2.5. If $d(L) < 0$, then L is a $(0, \dots, 0, a_r)$ -sheaf, with $r - 1$ zeros. For $r \geq 1$, $h^0(L(-1)) = 0$ by Lemma 2.4; then one cuts by integral hyperplane sections and uses Lemma 1.7.

Lemma 2.7. Assume $r = 2$ and $d(L) \leq 0$. Then, for every n ,

$$h^1(L(n)) \leq a_1(a_1 + 1)a_2 - a_0, \quad a_0 \leq a_1(a_1 + 1).$$

The proof cuts by an integral hyperplane section Y and uses the exact sequence $0 \rightarrow L(n-1) \rightarrow L(n) \rightarrow L_Y(n) \rightarrow 0$, Lemma 2.5, and the bounded-regularity estimate.

Lemma 2.8. Under the same hypotheses, with $d(L) < 0$, the estimate of Lemma 2.7 admits a uniform polynomial refinement in terms of the degree and Hilbert coefficients; in particular all $h^1(L(n))$ are bounded by a universal expression depending only on the Hilbert polynomial of L and on H .

Lemma 2.9. Assume $r \geq 2$, $d(L) < 0$, and X normal. Then:

- i) $H^0(L(-n)) = 0$ for $n \geq 2 + h^1(\omega_X \otimes L^{-1})$;
- ii) the numbers $h^1(L(n))$ are bounded, for all n , by universal polynomials in the last Hilbert coefficients of L and the fixed polarization;
- iii) there are universal polynomials β_i with rational coefficients, independent of k and X , such that $|a_i| \leq \beta_i(a_0, \dots, a_r)$.

The proof is by induction on dimension, cutting by a normal hyperplane section Y and using the exact cohomology sequence for $0 \rightarrow L(-n-1) \rightarrow L(-n) \rightarrow L_Y(-n) \rightarrow 0$.

Remark 2.10. For a fixed integer m , the two expansions

$$\sum_{i=0}^r a_i^{(m)} \binom{n+i}{i} = \sum_{i=0}^r a_i \binom{n+m+i}{i}$$

show that the coefficients $a_i^{(m)}$ are universal polynomials in m and in the a_i .

Theorem 2.11. Let $\mathcal{L}$ be a family of classes of invertible sheaves on the fibres of a projective morphism X/S , with S noetherian. If the Hilbert polynomials of members of $\mathcal{L}$ form a finite set and the last coefficients satisfy the universal bounds supplied above, then $\mathcal{L}$ is bounded. In particular, for a fixed Hilbert polynomial, the corresponding classes of invertible sheaves form a bounded family.

3. Bounded families of invertible sheaves

Lemma 3.1. Let S be a scheme and X an S -scheme. A family of invertible sheaves on geometric fibres which is represented by an invertible sheaf on X_T , for T of finite type over S , remains bounded after arbitrary base change. Conversely, a boundedness statement may be checked after replacing S by finitely many locally closed subschemes.

Lemma 3.4. Boundedness of families of classes of invertible sheaves is stable under finite faithfully flat base change, and under passage to a finite family of components. More explicitly, if $S' \rightarrow S$ is finite faithfully flat and a family becomes bounded over S' , then it was bounded over S ; and if $X = \coprod X_i$, boundedness on the fibres of X/S is equivalent to boundedness on the fibres of each X_i/S .

Lemma 3.5. Let $f : X' \rightarrow X$ be finite and surjective, with X and X' proper over S . A family of classes of invertible sheaves on geometric fibres of X/S is bounded if and only if its inverse-image family under f is bounded.

Theorem 3.6. Let S be noetherian and X proper over S . A family of classes of invertible sheaves on geometric fibres of X/S is bounded if and only if, after suitable reductions by Chow presentations, finite flat base change, and passage to integral projective models, it is bounded in the projective integral case.

In the projective integral case, with a very ample sheaf $\mathcal{O}_X(1)$, a bounded family $\mathcal{L}$ of classes has Hilbert polynomials constant in algebraic families: for an invertible sheaf L on X_T , flat over connected T , the polynomial

$$P(q, m) = \chi(L_t^{\otimes q}(m))$$

is independent of t . Conversely, if the classes $M_t^{\otimes n}$ lie in a bounded family, comparison of the polynomials

$$\chi(M_t^{\otimes q}(m)) = \sum_i a_i(q) \binom{m+i}{i}$$

shows that the $a_i(q)$ are independent of the chosen member; hence the family has a single Hilbert polynomial and is bounded by Theorem 2.11.

Theorem 3.8. Let S be noetherian. Let Y be projective over S , with relatively ample invertible sheaf $\mathcal{O}_Y(1)$, and let X be the zero-scheme of a section of $\mathcal{O}_Y(1)$. If, for every geometric fibre and every irreducible component of the fibre of Y , all irreducible components of the intersection with X have dimension at least 2, then

$$f^* : \text{Pic}_{Y/S} \longrightarrow \text{Pic}_{X/S}$$

is of finite type. If S is integral, after restricting to a non-empty open $U \subset S$, the Picard functors are representable and the induced morphism over U is quasi-affine, hence of finite type.

The proof reduces by noetherian induction, finite flat base change, and normalization to the case where S is integral and Y is normal and integral over S . If M_s is an invertible sheaf on a geometric fibre Y_s , and

$$\chi(M_s(n)) = \sum_i a_i \binom{n+i}{i},$$

then the Hilbert polynomial of $M_s|_{X_s}$ is obtained by shifting the coefficients. The boundedness criterion of Theorem 1.13 and the estimates of Section 2 give finite type. The generic representability and quasi-affineness follow from Exposé XII and from the finiteness of the kernel in the following lemma.

Lemma 3.9. If S is integral noetherian and X is integral over S , then after replacing S by a non-empty open and then by a finite faithfully flat integral cover S' , there exists a normal integral S' -scheme X' and a finite surjective S' -morphism

$$X' \longrightarrow X_{S'}.$$

Remark 3.10. If a resolving morphism exists generically, the same argument gives, after such a base change, a scheme X' smooth over S' and a proper surjective birational morphism $X' \rightarrow X_{S'}$.

Lemma 3.11. Let k be algebraically closed. Let Y be a normal projective k -scheme with ample sheaf $\mathcal{O}_Y(1)$, and let X be the zero-scheme of a section of $\mathcal{O}_Y(1)$. If $\dim Y \geq 3$, then $\text{Pic}_Y \rightarrow \text{Pic}_X$ is of finite type and has finite kernel. If $\dim Y \geq 2$, the same conclusion holds for $\text{Pic}_Y^0 \rightarrow \text{Pic}_X^0$.

The kernel is finite because, by Kummer theory,

$$0 \rightarrow \mu_\ell \rightarrow \mathbb{G}_m \xrightarrow{\ell} \mathbb{G}_m \rightarrow 0$$

and Hilbert's Theorem 90, a non-trivial subgroup of type $\mathbb{Z}/\ell\mathbb{Z}$ would give a non-trivial element killed by the map $H^1(Y, \mathbb{Z}/\ell\mathbb{Z}) \rightarrow H^1(X, \mathbb{Z}/\ell\mathbb{Z})$. But this map is injective by the surjectivity of $\pi_1(X) \rightarrow \pi_1(Y)$, equivalently by the connectedness theorem for the restriction of connected étale covers.

Remark 3.12. Under the hypotheses of 3.11, the kernel is a finite unipotent group. In characteristic zero it is therefore zero, recovering in this form the corresponding Kodaira-vanishing statement in Mumford's formulation.

Theorem 3.13. Let S be noetherian and let X be projective over S , with relatively ample sheaf $H = \mathcal{O}_X(1)$. Assume the geometric fibres are integral of dimension r . For a family $\mathcal{L}$ of classes of invertible sheaves on the fibres of X/S , the following conditions are equivalent:

- i) $\mathcal{L}$ is bounded;
- ii) in $\chi(L_s(n)) = \sum_{i=0}^r a_i(s) \binom{n+i}{i}$, the coefficient a_{r-1} remains bounded and a_{r-2} remains bounded below;
- iii) the degree $d(L_s) = \langle c_1(L_s) c_1(H_s)^{r-1} \rangle$ remains bounded, and $\langle c_1(L_s)^2 c_1(H_s)^{r-2} \rangle$ remains bounded below;
- iv) there is an integer N such that $L^{\otimes N} < L < L^{\otimes(-N)}$ in the order relation defined by ampleness, i.e. $L_s^{\otimes N} \otimes H_s^{-1}$ and $L_s^{\otimes(-N)} \otimes H_s$ are ample for every L_s in the family.

The implications from boundedness use EGA II, 4.6.12 and the constancy of $\chi(L_t^{\otimes m}(n))$ in flat connected families. Condition (iv) gives the indicated degree inequalities by positivity. Conversely, by replacing H by a multiple, cutting down to a surface, and using Theorem 3.8 and the estimates of Section 2, the boundedness follows.

4. Finiteness theorems for $\text{Pic}_{X/S}^\tau$

4.1. For a functor G in commutative groups over S , let G^0 and G^τ be the subgroup functors defined as follows. A T -point $f : T \rightarrow G$ lies in G^0 if, for every algebraically closed field K and every K -point t of T , the point $f(t)$ can be joined to 0 by a connected finite-type K -scheme mapping to G . It lies in G^τ if some non-zero multiple of $f(t)$ lies in $G^0(K)$. The formation of G^0 and G^τ commutes with base change. If S is a field and G is represented by a locally finite type group scheme, G^0 is represented by the connected component of the identity.

Lemma 4.2. If $\Phi : G \rightarrow G'$ is a morphism of commutative group functors such that, on every algebraically closed fibre, G and G' are represented by locally finite type group schemes and Φ is of finite type, then

$$\Phi(G^\tau) \subset G'^\tau.$$

Indeed, membership in $G^\tau(K)$ is equivalent to quasi-compactness of the sequence $(ng)_{n \in \mathbb{N}}$, and quasi-compactness is preserved by Φ .

Lemma 4.3. Let X and Y be proper over S , and let $f : X \rightarrow Y$ be either a surjective morphism or the inclusion of the zero-scheme of a section of an ample sheaf $\mathcal{O}_Y(1)$. Then

$$(f^*)^{-1}(\text{Pic}_{Y/S}^\tau) = \text{Pic}_{X/S}^\tau.$$

This follows formally from 4.2, XII 1.2, 3.1, and 3.5, respectively 3.8.

4.4. Let k be algebraically closed, X proper over k , and L an invertible sheaf on X . We say that L is algebraically equivalent to zero if there exist a connected finite-type k -scheme T , an invertible sheaf M on $X \times_k T$, and points $t, t_0 \in T(k)$ with

$$L \simeq M_t, \quad \mathcal{O}_X \simeq M_{t_0}.$$

It is τ -equivalent to zero if some non-zero tensor power is algebraically equivalent to zero. Equivalently, the corresponding point of $\text{Pic}_{X/k}$ lies in $\text{Pic}_{X/k}^0(k)$, respectively in $\text{Pic}_{X/k}^\tau(k)$. We also recall that L is numerically equivalent to zero if

$$\langle c_1(L) \cdot Y \rangle = 0$$

for every closed integral curve Y in X .

Proposition 4.5. Let $f : X' \rightarrow X$ be a morphism of proper k -schemes and let L be an invertible sheaf on X .

- i) If L is τ -equivalent to zero, respectively numerically equivalent to zero, then f^*L is so.
- ii) Conversely, if f is surjective and f^*L is τ -equivalent to zero, respectively numerically equivalent to zero, then L is so.

For numerical equivalence, if Y' is an integral curve in X' and $Y = f(Y')$, then, when $\dim Y = 1$,

$$\langle c_1(f^*L) \cdot Y' \rangle = [k(Y') : k(Y)] \langle c_1(L) \cdot Y \rangle.$$

When $\dim Y = 0$, $(f^*L)|_{Y'} \simeq \mathcal{O}_{Y'}$. Conversely, for a curve $Y \subset X$, choose a curve $Y' \subset X'$ dominating it and apply the displayed formula.

Theorem 4.6. Let k be algebraically closed, X proper over k , and L an invertible sheaf on X . The following are equivalent:

- i) L is τ -equivalent to zero;
- ii) for every coherent sheaf F on X , $\chi(F \otimes L) = \chi(F)$;
- iii) L is numerically equivalent to zero.

If X is projective over k , with ample invertible sheaf $H = \mathcal{O}_X(1)$, these are also equivalent to:

- iv) for every integer m , positive or negative, $L^{\otimes m} \otimes H$ is ample;
- v) if X is irreducible, then for all integers m, n ,

$$\chi(L^{\otimes m}(n)) = \chi(\mathcal{O}_X(n));$$

- vi) if X is irreducible of dimension $r \geq 2$,

$$\langle c_1(L)c_1(H)^{r-1} \rangle = 0, \quad \langle c_1(L)^2c_1(H)^{r-2} \rangle = 0.$$

To prove (i) $\Rightarrow$ (ii), dévissage and Proposition 4.5 reduce to X integral and $F = \mathcal{O}_X$; then L may be replaced by a non-zero tensor power and deformed to $\mathcal{O}_X$ in a connected algebraic family, so $\chi(L^{\otimes m})$ is constant. Applying (ii) to $F = \mathcal{O}_Y$ for an integral curve Y gives (iii) by Riemann's theorem for curves. A Chow presentation reduces (iii) $\Rightarrow$ (i) to the projective case. There, passing to reduced irreducible components reduces to X integral. If X is irreducible, the implications (ii) $\Rightarrow$ (v), (iii) $\Rightarrow$ (vi), and (v) $\Rightarrow$ (vi) are immediate from 2.1; (vi) is unchanged on X_{red} . Under (vi), respectively (iv), the family $\{L^{\otimes m}\}_{m \in \mathbb{N}}$ satisfies Theorem 3.13 and is bounded. Two distinct powers then lie in the same connected component of a finite-type parameter space, so a non-zero tensor power of L is algebraically equivalent to zero. Thus (vi), respectively (iv), implies (i), and (vi) also gives (iv).

Theorem 4.7. Let $f : X \rightarrow S$ be proper and flat, with S noetherian. Then:

- i) the subfunctor $\text{Pic}_{X/S}^\tau$ of $\text{Pic}_{X/S}$ is represented by an open subscheme of $\text{Pic}_{X/S}$;
- ii) if, in addition, X is projective and integral over S , then $\text{Pic}_{X/S}^\tau \rightarrow \text{Pic}_{X/S}$ is represented by a closed immersion;
- iii) $\text{Pic}_{X/S}^\tau$ is of finite type over S , equivalently, the family of classes of invertible sheaves τ -equivalent to zero on the fibres is bounded.

For a morphism $T \rightarrow \text{Pic}_{X/S}$, the question descends along the limit expression of an affine T as a filtered limit of finite-type S -schemes; after f.p.p.f. descent, it is enough to study a section defined by an invertible sheaf L on X . Let S_0 be the set of $s \in S$ for which L_s is τ -equivalent to zero. One must prove that S_0 is open, and under (ii) also closed. This is reduced to the following three assertions: generalizations of points of S_0 remain in S_0 ; specializations sufficiently close to a point of S_0 remain in S_0 ; and, in the projective integral case, all specializations of a point of S_0 remain in S_0 .

Using a Chow presentation $X' \rightarrow X$, it suffices, by 4.3 and 3.5, to treat the projective case. If H is ample for X/S and $(L^{\otimes m} \otimes H)_s$ is ample, the same is true at every generalization t of s ; by 4.6 this proves the first assertion. For the remaining assertions and for (iii), reduce by finite

flat base change and noetherian induction to the case where S is integral and X is projective and integral over S . Then, for a very ample $\mathcal{O}_X(1)$ and all m, n , the polynomials

$$\chi(L_t^{\otimes m}(n)), \quad \chi(\mathcal{O}_{X_t}(n))$$

are independent of $t \in S$. Hence specialization follows from 4.6 (i) $\Leftrightarrow$ (v), and the family of τ -equivalent invertible sheaves on the fibres has a single Hilbert polynomial; (iii) follows from Theorem 2.11.

Remark 4.8. If X is proper and integral over a noetherian scheme S , without projectivity over S it is not known whether $\text{Pic}_{X/S}^\tau \rightarrow \text{Pic}_{X/S}$ is still a closed immersion, except when X is normal over S and $\text{Pic}_{X/S}$ is representable.

5. Finiteness theorems of Néron–Severi type

Theorem 5.1. Let X be proper over a noetherian scheme S . Then the Néron–Severi groups

$$\text{NS}(X_K) = \text{Pic}_{X_K/K}(K) / \text{Pic}_{X_K/K}^0(K)$$

of the geometric fibres X_K of X/S are finitely generated, and their ranks and the orders of their torsion subgroups remain bounded.

By noetherian induction, it is enough to replace S by an arbitrary non-empty open of S_{red} , so that $\text{Pic}_{X/S}^\tau$ is represented by an S -scheme of finite type. Then the order of

$$\text{Tors}(\text{NS}(X_K)) = \text{Pic}_{X_K/K}^\tau(K) / \text{Pic}_{X_K/K}^0(K)$$

is the number of connected components of $(\text{Pic}_{X/S}^\tau)_K$, a number constant above a non-empty open of S . It remains to bound the rank of $\text{Pic}_{X_K/K}(K) / \text{Pic}_{X_K/K}^\tau(K)$. As in the proof of 4.7(iii), one reduces to S integral and X projective and integral over S . If r is the common dimension of the fibres, the rank is plainly 0 for $r = 0$ and 1 for $r = 1$. For $r \geq 2$, after shrinking S one chooses a linear section Y of X which is integral over S and has fibres of dimension 2; the proof continues from this reduction.